Sustainability in the AI Policies of Open Source Projects: A Grey Literature Review

Idea note for ICT4S 2027

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Motivation

- Open source software (OSS) projects increasingly rely on generative AI (GenAI): coding agents were adopted by 22–29% of 128,018 GitHub projects within months of their release (Robbes et al. 2026).
- Practices are highly diverse: no convention exists, even on what contributors must disclose, and many policy files are revised within months (Hora et al. 2026).
- Policies are written mostly in reaction to GenAI ("AI slop", review load, licensing), not out of a sustainability agenda (Hora et al. 2026).
- Sustainability is a *wicked problem* (Rittel and Webber 1973): its dimensions are interdependent, improving one can degrade another, and no single objective can be optimised.
- We do not know whether, and how, OSS communities frame sustainability when they regulate GenAI.

Research questions and hypotheses

- **RQ1.** Is sustainability addressed in the stated purpose (*raison d'être*) of open source projects?
  - *H1:* yes, or partially.
- **RQ2.** Is sustainability addressed in the AI policies of open source projects?
  - *H2:* no, or only partially.
- **RQ3.** Which definitions and characteristics of sustainability are addressed in these AI policies?
  - *H3:* mostly technical and individual sustainability, centred on the maintainer (workload, code quality, burnout).
- **RQ4.** Do these definitions provide frameworks leading to strong sustainability?
  - *H4:* no.

Methodology

We follow the guidelines for multivocal and grey literature reviews of Garousi et al. (2019). AI policies are grey literature (`CONTRIBUTING.md`, `AI_POLICY.md`, governance pages, discussion threads), so a classical systematic literature review does not apply.

Planning

- Define the review goal and the four RQs above.
- Decide that grey literature is the primary source, because academic work does not yet cover policies produced by communities.
- Complement it with a short academic search to ground the sustainability framework used in the coding scheme.

Source selection and search

- **Project sample.** Select OSS projects by explicit criteria (for example popularity, foundation membership such as Linux Foundation or Apache, domain diversity), using the corpus of Hora et al. (2026) as a starting point.
- **Policy files.** Collect AI policies from the repositories, project websites, and governance documents.
- **Purpose statements (RQ1).** Collect README, mission statements, manifestos, and charters.
- **Search.** Keyword queries (e.g., "AI policy", "LLM", "Copilot", "generated code") on GitHub code search and web search engines, with snowballing from links and mentions.
- **Stopping criterion.** Stop when new sources no longer add new codes (theoretical saturation).
- **Freezing.** Fix a collection date and archive every document (Software Heritage, Wayback Machine) for reproducibility, since policies change quickly.

Inclusion and quality assessment

- **Inclusion.** The document states rules or positions on AI-generated contributions or AI tool use in the project.
- **Exclusion.** Duplicates, forks that copy a policy verbatim (counted once), documents not in English or French, and unofficial opinions of individuals.
- **Quality.** Apply the grey literature criteria of Garousi et al. (2019): authority of the producer, methodology, objectivity, date, novelty, and impact.
- Record whether each document is official (merged policy) or a proposal under discussion.

Data extraction

- Project metadata: size, age, governance, foundation, domain.
- Policy metadata: date, version history, stance (ban, permit with conditions, permit freely), disclosure requirement.
- Verbatim excerpts that mention sustainability or related terms.

Coding scheme and synthesis

- **Sustainability dimensions.** Deductive codes from the Karlskrona manifesto (Becker et al. 2015): individual, social, economic, environmental, and technical. Inductive codes are added as they emerge.
- **Strength of sustainability.** Each definition is classified as *weak* (substitutability between natural and other capital) or *strong* (critical natural capital must be preserved) (Neumayer 2013).

- **Procedure.** Thematic analysis (Braun and Clarke 2006), with two coders on a sample, inter-rater agreement (Cohen's kappa), and disagreements resolved by discussion.
- **Mapping to RQs.**
  - RQ1: presence or absence of sustainability in purpose statements.
  - RQ2: presence or absence of sustainability in AI policies.
  - RQ3: distribution of dimensions and definitions across policies.
  - RQ4: classification of definitions as weak or strong, and analysis of interdependencies between dimensions.

Threats to validity

- **Selection bias.** Popular projects may not represent the long tail; mitigated by explicit sampling criteria and a stratified sample.
- **Volatility.** Policies change; mitigated by a frozen and archived corpus.
- **Implicit sustainability.** Projects may address sustainability without naming it; mitigated by a broad keyword set and manual reading.
- **Coder subjectivity.** Mitigated by a double-coded sample and agreement metrics.
- **Policy versus practice.** Policies say little about actual behaviour; this is a stated limit of the study.

Expected contributions

- A map of how OSS communities frame sustainability when regulating GenAI.
- A coding grid of sustainability dimensions that can be reused.
- Recommendations for maintainers, for instance a template sustainability clause for AI policies.

References

Becker, Christoph et al. 2015. "Sustainability Design and Software: The Karlskrona Manifesto." *Proc. ICSE*.

Braun, Virginia, and Victoria Clarke. 2006. "Using Thematic Analysis in Psychology." *Qualitative Research in Psychology* 3 (2): 77–101.

Garousi, Vahid, Michael Felderer, and Mika V. Mäntylä. 2019. "Guidelines for Including Grey Literature and Conducting Multivocal Literature Reviews in Software Engineering." *Information and Software Technology* 106: 101–21.

Hora, Andre, Romain Robbes, and Stefano Zacchiroli. 2026. *"We Permit the Use of AI, but [...]": The Landscape of AI Policies in Popular Open Source Projects.*

Neumayer, Eric. 2013. *Weak Versus Strong Sustainability.* 4th ed. Edward Elgar.

Rittel, Horst W. J., and Melvin M. Webber. 1973. "Dilemmas in a General Theory of Planning." *Policy Sciences* 4 (2): 155–69.

Robbes, Romain et al. 2026. *Agentic Much? Adoption of Coding Agents on GitHub.*